Few-Shot Hyperspectral Image Classification With Self-Supervised Learning

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***Abstract*— Recently, few-shot learning (FSL) has been introduced for hyperspectral image (HSI) classification with few labeled samples. However, existing FSL-based HSI classification methods mainly focus on the meta-knowledge transfer between HSIs. Compared with HSIs, natural images have sufficient annotated data. To utilize natural images (base class data) to achieve accurate classification of HSIs (novel class data), we propose a novel few-shot classification framework with SSL (FSCF-SSL) for HSIs in this article. The orientation of objects in natural images is relatively unitary, whereas the objects of image patches for each pixel in HSIs have diverse orientations in the spatial domain. To make better use of base classes, we design an SSL with geometric transformations (SSLGTs), which sets rotation labels as supervision to extract low-level features that can better represent diverse orientations, and then conduct SSLGT and FSL on base classes to learn transferable spatial meta-knowledge. Next, a spectral-spatial feature extraction network is carefully designed to better utilize the spatial and spectral information of HSIs, where the weights of the first seven layers of the spatial part are initialized by the weights of the corresponding layers trained on base classes. Finally, to fully explore the few annotated data from novel classes, we design an SSL with contrastive learning (SSLCL) that can mine the category-invariant features contained in the novel class data itself, and then perform SSLCL and FSL on novel classes to learn more discriminative individual knowledge. Experimental results on four HSI datasets show that FSCF-SSL offers a significant improvement over state-of-the-art methods. The code is available at https://github.com/Li-ZK/FSCF-SSL-2023.**

***Index Terms*— Contrastive learning (CL), few-shot learning (FSL), hyperspectral image (HSI) classification, self-supervised learning (SSL).**

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I. Introduction

HYPERSPECTRAL image (HSI) is a 3-D image captured by an airborne platform equipped with a hyperspectral imager. Each pixel in HSI contains hundreds of bands of solar reflection, which makes it useful for many practical applications, such as environmental monitoring, military target detection, and precision agriculture [1].

HSI classification is used to determine the corresponding object class for each pixel, which is important for HSI analysis and applications. Traditional HSI classification methods mainly focus on manual feature extraction [2], [3], [4] for traditional classifiers [5], [6], [7]. To further improve the performance of HSI classification, researchers have proposed a series of optimization methods based on traditional classification methods, such as kernel function [8], and sparse representation [9]. With the rapid development of deep learning technology, HSI classification based on deep learning has become a research hotspot. Chen et al. proposed a deep learning framework based on the stacked auto-encoder (SAE). The objective is to obtain effective high-level features for HSI classification [10]. Then, Chen et al. [11] used a deep belief network (DBN) to extract features in HSIs by means of spectral-spatial feature extraction and complete classification by logistic regression. The above two methods use fully connected (FC) layers as the network, leading to many network parameters and requiring diverse training data. To reduce network parameters, a convolutional neural network (CNN) is applied to HSI classification [12]. Chen et al. [13] proposed a CNN-based 3-D feature extraction model, which uses regularization to extract effective spectral-spatial features. Li et al. [14] proposed a pixel pair method to increase the number of samples to ensure that the network is fully trained, and the final classification is achieved by voting. The method proposed by Yang et al. [15] uses spatial background and spectral correlation to augment the classification of HSIs. Li et al. [16] designed an HSI classification framework (3DCNN) based on 3-D CNN. Zhong et al. [17] proposed an end-to-end spectral-spatial residual network (SSRN), in which spectral and spatial residual blocks can learn discriminative features. Yang et al. [18] proposed a multilevel spectral-spatial transformer network (MSTNet) for HSI classification. Zhou et al. [19] proposed a capsule network based on spatial-spectral joint representation (JSSR-CapsNet) to mine spatial-spectral information to improve classification performance. Yu et al. [20] proposed a dual-channel convolu-

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tional network (DCCN) for HSI classification to explore global and multiscale information on HSIs.

Although the above CNN-based HSI classification methods have achieved promising performance, training efficient and accurate CNN usually requires a large amount of labeled data. When the training data are very limited, CNN may suffer from overfitting, which affects the classification performance [4]. In addition, for hyperspectral remote sensing data, acquiring annotated data is time-consuming and expensive. Therefore, how to achieve higher classification accuracy with very few labeled samples has become an urgent problem to be solved [5].

To address the label scarcity challenge in deep learning [21], researchers have proposed various deep learning strategies with few labeled samples. Data augmentation is an intuitive idea, which uses a series of transformation operations to expand the number of labeled samples [22], [23]. In addition, some active learning methods [24], [25] have been applied to solve the problem with a few labeled samples.

Recently, inspired by the human visual system's ability to recognize novel classes with few labeled examples, few-shot learning (FSL) has attracted the attention of researchers [26], [27], [28], [29]. Meta-learning using the episodic learning method is one of the learning strategies of FSL, which includes meta-training and meta-testing phases [28]. In the meta-training phase, the training dataset is decomposed into meta-tasks containing different categories, and the model is trained with a large number of meta-tasks to obtain meta-knowledge that can be generalized to the classification of new unseen categories in the meta-testing phase. More recently, FSL using meta-learning strategies has been used for HSI classification [30], [31], [32], [33], [34]. Liu et al. [30] proposed a deep FSL (DFSL) method for HSI classification, which designs a deep 3-D residual network to learn a metric space that maximizes the discriminability and separability of classes. Li et al. [31] proposed a deep cross-domain FSL (DCFSL) approach for HSI classification, which addresses the problem of FSL and domain adaptation within a unified framework to learn a classification model that is not susceptible to domain shift. Wang et al. [32] proposed a heterogeneous FSL (HFSL) for HSI classification by learning transferable spatial structure and texture knowledge in natural images. Zhang et al. [33] proposed a graph information aggregation-based cross-domain FSL framework by combining FSL with graph information aggregation-based domain alignment. Yu et al. [34] proposed an edge-inferring graph neural network with dynamic task-guided self-diagnosis for few-shot HSI classification. Zuo et al. [35] proposed a few-shot classification method for HSIs based on an edge-labeling graph neural network.

Although FSL methods for HSIs classification have achieved good performance, they still cannot solve the problem of lack of labeled samples very well. On one hand, most current FSL-based HSI classification methods focus on meta-knowledge transfer between HSIs. Compared with HSIs, which are difficult to obtain more annotated samples, natural images have sufficient annotated data, and making full use of natural scene images to classify HSIs may alleviate the problem of insufficient samples to some extent. However, the orientation of objects in natural images is relatively unitary, while the objects of image patches for each pixel in HSIs have diverse orientations in the spatial domain (see Fig. 1), which poses a challenge to solving this problem. Therefore, it is necessary to extract low-level features from natural images that can better represent different orientations, so as to better promote the learning of spatial meta-knowledge suitable for HSIs. On the other hand, current HSI classification methods based on FSL directly use the few annotated data from novel classes to train models, which is not sufficient to mine the intrinsic representations of these data. Mining the intrinsic representations of the few annotated data itself is important to help learn more discriminating individual knowledge from novel classes. Recently, self-supervised learning (SSL) has been used to address the problem of insufficient labeled samples and has made significant progress [36]. SSL aims to enable the model to learn a richer image representation from the data itself. For instance, some super pixel-based SSL methods have been designed for HSI classification, which takes full advantage of each labeled sample itself [37], [38].

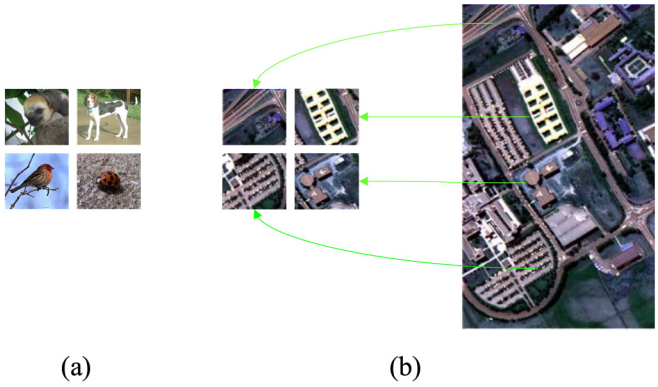

Fig. 1. Two different types of images. (a) Natural images from the mini-ImageNet dataset. The orientation of the objects is relatively unitary. (b) Image patches and the corresponding HSI dataset. The objects of image patch for each pixel in HSIs have diverse orientations.

To make full use of the abundant annotated data in natural images and the few annotated data in HSIs, we propose a novel few-shot classification framework with SSL (FSCF-SSL) for HSIs in this article, where natural images are regarded as base class data and HSIs are referred to as a novel class data. Specifically, to ensure spatial consistency, the spatial domain of HSI is mapped to the same dimension as the natural image through the mapping layer. First, to learn transferable spatial meta-knowledge more suitable for novel classes, we design an SSL with geometric transformations (SSLGTs) and conduct SSLGT and FSL on base classes. SSLGT uses multiple GTs to generate rotated images at different angles and sets rotation labels as supervision to extract low-level features that can better represent diverse orientations from base classes. Then, to make the best use of spatial and spectral information, we design a spectral-spatial network, which uses the residual network with pure 1-D operators to obtain spectral information, and the convolutional network with pure 2-D operators to extract the spectral information. The weights of

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the first seven layers of the spatial network are initialized by the weights of the corresponding layers trained on base classes. Finally, to learn more discriminative individual knowledge from novel classes, we devise an SSL with contrastive learning (SSLCL) and perform SSLCL and FSL on novel classes. SSLCL employs a spectral-spatial network terminated by a softmax operation to produce the class distributions of the two augmented features generated by a dropout operation, then sharpens the class distribution by minimizing the entropy of the class distribution for each data, and diversifies the prediction results across samples by maximizing the entropy of the mean class distribution, thus allowing high-quality category-invariant features contained in the data itself to be learned from novel classes. Experimental results on four HSI datasets show that our FSCF-SSL outperforms existing state-of-the-art deep learning and FSL methods. The main contributions of this article are summarized as follows:

1) An SSLGT strategy with GTs is executed on base classes, which can extract low-level features that can better represent diverse orientations from abundant natural images.

2) An SSLCL strategy with CL is implemented on novel classes, which can extract high-quality category-invariant features from a few labeled HSIs.

3) FSL and SSL are employed to train the model to realize knowledge extraction, which not only learns the transferable spatial meta-knowledge suitable for HSIs, but also learns the more discriminative individual knowledge for HSIs.

The rest of this article is structured as follows. Section II briefly introduces the related works. Section III details our proposed FSCF-SSL method. The experimental results and detailed analysis are shown in Section IV. Finally, conclusions are drawn in Section V.

II. Related Work

A. Few-Shot Learning

The goal of FSL is to bridge the gap between humans and machines by enabling machine learning models to identify novel categories from a few samples, similar to humans. A core idea of FSL is to acquire knowledge by training a model on a dataset with a large number of labeled samples, and then use this knowledge to train the model, so that the model can be better extended to the task of identifying novel categories from a few samples [39]. In general, FSL is trained by constructing a task consisting of a support set and a query set. Assume N classes are randomly selected from a training set and K samples are selected in each class. Then the selected $N \times K$ samples are called the support set and some data from the remaining samples of the N categories are selected to form the query set. FSL consists mainly of meta-learning-based and metric-based approaches.

The meta-learning-based method uses meta-learning to train a meta-learner for FSL. Ravi and Larochelle [40] adopted a long short-term memory (LSTM)-based meta-learner as an optimizer to train another classifier. Cai et al. [41] proposed a novel memory-matching network (MM-Net), which follows the philosophy that training and testing conditions must be matched. Santoro et al. [42] proposed a meta-learning algorithm with memory enhanced neural network (MANN) for FSL.

The metrics-based approach is to transform the data into another low-level dimension and then classify it according to the distance between these embedding features. Snell et al. [43] proposed a prototype network for few-shot classification, which learns a metric space by taking the mean of each class as its corresponding prototype representation. Sung et al. [44] proposed a relational network to learn a deep nonlinear metric. Vinyals et al. [26] proposed a matching network combining attention and memory for FSL.

B. Self-Supervised Learning

The performance of deep neural networks depends on the amount of training data and collecting and annotating large-scale datasets can be time-consuming and expensive. SSL is a method of learning visual features from large-scale unlabeled images or videos, which avoids the need for time-consuming and expensive data annotation. As a promising paradigm for learning useful image representations, SSL utilizes the input data itself as supervision and facilitates almost all types of downstream tasks [36]. Such methods for image processing mainly include generation-based methods, context-based methods, and contrastive-based methods.

The generation-based methods involve the process of generating images, including image inpainting [45], image super-resolution [46], and image colorization [47]. Pathak et. al. [45] used a CNN to solve the image inpainting problem and employed adversarial loss to assist the network in generating a realistic hypothesis. Ledig et al. [46] proposed a generative adversarial network (GAN) for single-image super-resolution, which can recover photo-realistic textures from heavily down-sampled images and show significant improvements in perceived quality by using adversarial loss and content loss. Zhang et al. [47] used the image coloring task as a pretext for SSL, and the features learned through the colorization process were evaluated on other downstream high-level tasks after the image colorization training was completed.

The context-based methods primarily employ contextual features of images, including contextual similarity, spatial structure, and temporal structure, as supervisory signals. Doersch et al. [48] used spatial context as the source of supervision signals to learn a plentiful image representation. They extracted random patch pairs from each image to train a convolutional network that could predict the position of the second patch relative to the first. Noroozi and Favaro [49] introduced a context-free network that is trained by using a jigsaw puzzle as a pretext task to learn to recognize each tile as a piece of an object, and how these pieces fit together in the object. Gidaris et al. [50] proposed a self-supervised method for predicting the rotation angle of an image. Each image is rotated at four angles by performing multiple GTs, and then the network predicts the rotation angle of the image.

The contrastive-based methods learn an embedded space in which different augmentations features of the same image are attracted while features of different images are separated. Chen and He [51] proposed solutions to abandon negative samples and designed special techniques such as asymmetric structures, momentum encoders, and stop gradients to avoid

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collapse. Zbontar et al. [52] proposed learning information representations by reducing redundancy or covariance in different representational dimensions. Wang et al. [53] proposed an SSL framework without using negative samples to solve the collapse problem during model training by maximizing the mutual information between input and output.

III. Proposed FSCF-SSL Method

To better explain the proposed FSCF-SSL for HSI classification, several important concepts are first introduced. FSL involves base class data D_s and novel class data D_t. In our approach, mini-ImageNet is used as the base class data and the HSI dataset is treated as the novel class data. D_t can be further divided into two parts: D_l with few labeled samples for training and D_u without labeled samples for testing, and $D_l \cup D_u = D_t$. Notably, there are very few labeled samples in D_l, which cannot generate a training episode. Therefore, data augmentation is used to increase the number of samples in each class to 200. Specifically, random Gaussian noise [54] is added to D_l for data augmentation. Given a known training sample x, the new virtual sample x' can be generated by

$$x' = f(x) + \gamma n \tag{1}$$

where $f(.)$ represents a transformation function and γ controls the weight of the random Gaussian noise n. In general, D_s and D_l are used for training on base and novel classes, respectively, while D_u is used for testing novel classes. The notations used are shown in Table I.

TABLE I
Notation Used in FSCF-SSL

Notation	Definition	comment
D_s	Base class data	Base classes
S_s	Base class support set	
Q_s	Base class query set	
G_r	Rotated samples for SSLGT	
N_s	The number of classes of each episode in the base class FSL	
P_s	The prototype calculated by the base class support set	
$x_s^{sup,i}, y_s^{sup,i}$	The ith sample and the corresponding label in the base class support set	
x_s^{que}, y_s^{que}	A sample and corresponding label in the base class query set	
L_s^{fsl}	Loss of base class FSL	
L_s^{gt}	Loss of SSLGT	
L_s^{total}	Total loss of base class training	
F_θ	The feature extraction of the base class with parameter θ	
x_s	A sample in the base class data	
$R_\emptyset$	The network with parameter $\emptyset$ used to predict the rotation category of the rotated sample	
B	Number of HSI bands	Novel classes
D_l	Novel class labeled data	
D_u	Novel class unlabeled data	
D_t	Novel class data	
P_t	The prototype calculated by the novel class support set	
x_t^{que}, y_t^{que}	A sample and corresponding label in the novel class query set	
C_l	Contrast set for SSLCL	
$(x_t^{C_l,1}, x_t^{C_l,2})$	A sample pair in C_l	
S_t	Novel class support set	
Q_t	Novel class query set	
L_t^{total}	Total loss of novel class training	
L_t^{fsl}	Loss of novel class FSL	
L_t^{cl}	Loss of SSLCL	
N_t	The number of classes of each episode in the novel class FSL	

A. FSCF-SSL Framework

For FSL on base classes, samples are randomly selected from D_s to form a support set and a query set, and the model is trained by a metric-based meta-learning method, which aims to bring samples of the same category closer together and samples of different categories further apart. For SSLGT on base classes, randomly selected samples from D_s are used to construct rotating samples (G_r) for model training, so that the model can extract low-level features that can better represent diverse orientations. SSLGT and FSL are executed on base classes to obtain the transferable meta-knowledge and extract low-level features from base classes that can better represent diverse orientations. Thus, the total loss for base classes is as follows:

$$L_s^{\text{total}} = L_s^{\text{fsl}} + L_s^{\text{gt}}. \tag{2}$$

For novel classes, a feature extraction model is used to extract the spectral-spatial fusion features. For FSL on novel classes, samples are randomly selected in D_l, and other settings are the same as FSL for base classes. For SSLCL on novel classes, samples are randomly selected from D_l to train the model. For each sample in D_l, we copy an identical sample to form a sample pair. Then, two identical features are obtained through the spectral-spatial extraction network. The dropout operation is performed on these two identical features. Due to the randomness of the dropout operation, two different features can be obtained. SSLCL and FSL are executed on novel classes to obtain high-quality category-invariant features and learn more discriminative individual knowledge. The total loss for novel classes is

$$L_t^{\text{total}} = L_t^{\text{fsl}} + L_t^{\text{cl}}. \tag{3}$$

Fig. 2 shows the procedure of the proposed FSCF-SSL framework. The framework first trains the model on base classes, then transfer some of the models trained on base classes to novel classes, and finally uses very few labeled samples in the novel class for fine-tuning. For base classes, training data for FSL and SSLGT are first constructed from the base class data (D_s), and then both FSL and SSLGT are performed on the model. For novel classes, samples are

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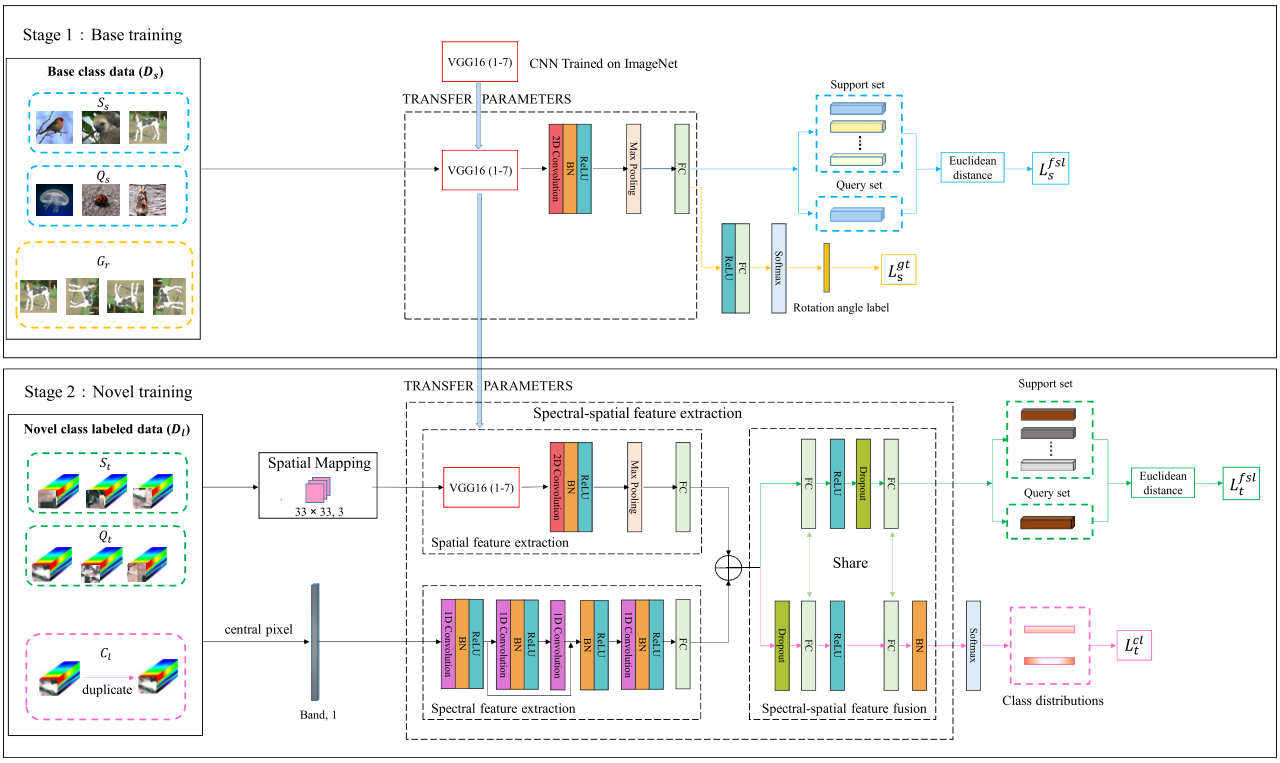

Fig. 2. Framework of proposed FSCF-SSL. The training on base classes and novel classes is performed sequentially. First, FSL and SSLGT are performed on the base class feature extraction to learn transferable spatial meta-knowledge and low-level features from base classes that can better represent diverse orientations. Then, FSL and SSLCL are performed on the spectral-spatial feature extraction to learn high-quality category-invariant features and more discriminative individual knowledge.

selected from the novel class labeled data (D_l) for FSL and SSLCL, and then the spectral-spatial feature extraction model is used to extract the fusion features of the samples. Finally, the extracted fusion features are used for FSL and SSLCL.

B. Spectral-Spatial Feature Extraction

Spectral-spatial embedded feature extraction consists of three modules: spectral feature extraction, spatial feature extraction, and spectral-spatial feature fusion.

1) Spatial Feature Extraction: The mapping layer proposed in [55] is used to deal with the channel mismatch between the samples of D_s and D_l. Specifically, a $33 \times 33 \times B$ image cube corresponding to the center pixel is mapped to a $33 \times 33 \times 3$ image cube, where B is the number of bands. The weights of the first seven layers of CNN trained on the D_s are transferred to the CNN model in HSI space. Then, spatial features are extracted using CNN with a 2-D convolution layer, batch normalization (BN) layer, rectified linear unit (ReLU) activation function, 2-D maximum pooling layer, and FC layer.

2) Spectral Feature Extraction: The network includes a 1-D convolution layer, spectral residual block, and FC layer. First, the input band vector passes through a 1-D convolutional layer to extract features. Then, in the spectral residual block, two 1-D convolution layers and an identity map are included to learn deep spectral features. Finally, feature vectors are sent to the last 1-D convolution layer and FC layer to generate spectral features.

3) Spectral-Spatial Feature Fusion: To make the best use of spatial and spectral information, we designed a spectral-spatial feature fusion model that includes two branches. The upper branch extracts spectral-spatial fusion features for FSL on novel classes, which consists of an FC layer, a ReLU, a dropout layer, and an FC layer. The lower branch extracts the spectral-spatial fusion features for SSLCL, which consists of a dropout layer, a ReLU, an FC layer, and a BN layer.

C. Base Class FSL

In the FSCF-SSL framework, the transferable base class knowledge can be learned by performing FSL on the base class data D_s. Specifically, N_s classes are randomly selected from D_s, and K labeled samples are randomly selected from each selected class to form a support set, which is defined as $S_s = \{(x_s^{\text{sup},i}, y_s^{\text{sup},i})\}_{i=1}^{K \times N_s}$, and then C samples are randomly selected to form a query set in each selected class, which is defined as $Q_s = \{(x_s^{\text{que},j}, y_s^{\text{que},j})\}_{j=1}^{C \times N_s}$. Samples from support and query sets do not intersect. The probability of the sample $x_s^{\text{que},j}$ in Q_s to belong to class m can be computed as

$$P\left(y_s^{\text{que},j} = m \mid x_s^{\text{que},j} \epsilon Q_s; \theta\right) = \frac{\exp\left(-d\left(F_\theta\left(x_s^{\text{que},j}\right), c_m\right)\right)}{\sum_{m=1}^{N_s} \exp\left(-d\left(F_\theta\left(x_s^{\text{que},j}\right), c_m\right)\right)} \tag{4}$$

where c_m is the embedded features of the mth class in the support set, $d(.)$ is the function of Euclidean distance, $x_s^{\text{que},j}$ and $y_s^{\text{que},j}$ are the jth sample and the corresponding label in the query set respectively, N_s is the number of classes for each episode, and Q_s is a query set generated by D_s.

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Finally, the classification loss for all query samples in an episode can be formulated as

$$L_s^{\text{fsl}} = E_{S_s,Q_s}\left[-\sum_{(x_s^{\text{que}},y_s^{\text{que}})\in Q_s}\log p\left(y_s^{\text{que}}=m\mid x_s^{\text{que}};\theta\right)\right]. \quad (5)$$

D. Self-Supervised Learning With Geometric Transformations

SSLGT on base classes aims to extract low-level features that can better represent diverse orientations, which can improve the ability of the spectral-spatial embedded feature extraction on novel classes to extract the spatial information of HSIs. First, four 2-D rotation transformations are defined $R = \{0°, 90°, 180°, 270°\}$, and then create four rotation copies of an image $\{x_s^r | r \in R\}$, where x_s^r is the image x_s rotated by r. Based on features $F_\theta(x_s^r)$ extracted from the rotated image, the new network $R_\emptyset$ tries to predict the rotation category. Accordingly, self-supervised loss is defined as

$$L_s^{\text{gt}}(D_s;\emptyset,\theta) = E_{x_s\sim D_s}\left[\sum_{\forall r\in R}-\log R_\emptyset^r\left(F_\theta\left(x_s^r\right)\right)\right] \quad (6)$$

where $R_\emptyset^r(.)$ is the predicted score for rotation r.

E. Novel Class FSL

FSL is performed on the novel class data D_l to extract discriminative features and individual knowledge, aiming to improve the performance of few-shot classification on D_u. First, N_t classes are randomly selected from D_l, then K labeled samples are selected from each class as the support set, and then C samples are selected from each class of the remaining samples as the query set. The training loss for each training episode can be defined as

$$L_t^{\text{fsl}} = E_{S_t,Q_t}\left[-\sum_{(x_t^{\text{que}},y_t^{\text{que}})\in Q_t}\log p\left(y_t^{\text{que}}=n|x_t^{\text{que}};\eta\right)\right] \quad (7)$$

where S_t is a support set generated by D_l and Q_t is a query set generated by D_l. η is the spectral-spatial feature extraction parameter without the last BN layer.

F. Self-Supervised Learning With Contrastive Learning

SSLCL can obtain high-quality category-invariant features to improve the classification accuracy for novel classes. A sample is randomly selected in D_l and duplicated to form a pair of identical samples $(x_t^{C_l,1}, x_t^{C_l,2})$. These two identical samples are then fed into the spectral-spatial feature extraction network to extract a pair of identical features. Dropout is then performed separately for the same pair of features. Due to the randomness of dropout, two augmented features can be obtained. The final output of SSL is two class distributions $z_t^{\text{cl},1}$ and $z_t^{\text{cl},2}$, and the self-supervised contrastive loss function [53] $L_t^{\text{cl}}(.)$ is defined as follows:

$$L_t^{\text{cl}}\left(z_t^{\text{cl},1}, z_t^{\text{cl},2};\gamma\right) = \frac{1}{2}\left(L\left(z_t^{\text{cl},1}||z_t^{\text{cl},2}\right) + L\left(z_t^{\text{cl},2}||z_t^{\text{cl},1}\right)\right. \quad (8)$$

where

$$L\left(z_t^{\text{cl},1}\|z_t^{\text{cl},2}\right) = \underbrace{\frac{1}{\text{Bs}}\sum_{i=1}^{\text{Bs}} D_{\text{kl}}\left(z_t^{\text{cl},i,1}\|z_t^{\text{cl},i,2}\right)}_{\text{consistency term}} + \underbrace{\frac{1}{\text{Bs}}\sum_{i=1}^{\text{Bs}} H\left(z_t^{\text{cl},i,1}\right)}_{\text{sharpness term}} - \underbrace{H\left(\frac{1}{\text{BS}}\sum_{i=1}^{\text{Bs}} z_t^{\text{cl},i,1}\right)}_{\text{diversity term}} \quad (9)$$

$L(z_t^{\text{cl},2}||z_t^{\text{cl},1})$ is defined in the same way, $D_{\text{kl}}(.||.)$ denotes the Kullback–Leibler divergence between two probability distributions, $H(.)$ denotes the entropy of a specific probability distribution, Bs denotes the batch size, and γ is the spectral-spatial feature extraction parameter. Specifically, minimizing the consistency term keeps the predictions of different features from the same sample consistent, that is, different features of the same image need to be identified as the same class. For the sharpness term, the output distribution is regularized to make it sharp by minimizing the class distribution entropy of each sample, which gives each sample a deterministic assignment. In addition, features assigned to the same class of samples become more compact, which can improve feature discrimination. For the diversity term, the entropy of the average distribution among different samples is maximized to make the prediction of different samples distributed on novel classes as far as possible, which can avoid the network assigning all images to the same class.

G. Implementation Details

In order to make full use of base classes to improve the classification performance of novel classes, the training process of the FSCF-SSL method consists of training on base classes and training on novel classes. For training on base classes, first, the samples for FSL and SSLGT are selected in D_s, and then FSL and SSLGT are performed on the base class feature extraction. Finally, the first seven layers of the VGG16 network in the base class feature extraction are transferred to the novel class spectral-spatial feature extraction. For novel class training, data for FSL and SSLCL are selected in D_l, and then FSL and SSLCL are performed on the spectral-spatial feature extraction. The training procedure is detailed in Algorithm 1.

IV. Experimental Results and Analysis

A. Experimental Dataset

To evaluate the performance of FSCF-SSL, we use the mini-ImageNet dataset as the base class data, and four HSI datasets as the novel class data, including the Indian Pines (IP), the University of Pavia (UP), the Salinas (SA), and the WHU-Hi-HanChuan (HC). Notably, the total number of samples in the WHU-Hi-HC dataset is too large, 15% of the samples in each class are randomly selected.

1) Base Class Data: Mini-ImageNet is a dataset with rich texture and structural information that is often used in the meta-learning domain. It contains 100 classes with a total of 60 000 color images. Each class has 600 samples, and each image is of size 84 × 84.

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Algorithm 1 FSCF-SSL

Input: The base class support set S_s and query set Q_s in D_s
The base class Rotated samples G_r in D_s
Base class training episodes E_s
The novel class support set S_t and query set Q_t in D_l
The novel class contrast set C_l in D_l
Novel class training episodes E_t
Predicting episodes E_{test}
Output: Each part outputs corresponding prediction labels respectively
1: for episode in $\{1, \ldots, E_s\}$ do
2: Train with S_s, Q_s and G_r
3: Calculate the prototype of $S_s \rightarrow P_s$
4: for $(x_s^{que}, y_s^{que}) \epsilon Q_s$ do
5: Calculate the L_s^{fsl} of an episode by Equation (5)
6: end for
7: for $x_s^r \epsilon G_r$ do
8: Calculate the L_s^{gt} of an episode by Equation (6)
9: end for
10: Return the L_s^{total} of an episode by Equation (2)
11: end for
12: Transfer the first 7 layers of the VGG16 network in the bass class feature extraction to the novel class spectral-spatial feature extraction
13: for episode in $\{1, \ldots, E_t\}$ do
14: Train with S_t, Q_t and C_l
15: Calculate the prototype of $S_t \rightarrow P_t$
16: for $(x_t^{que}, y_t^{que}) \epsilon Q_t$ do
17: Calculate the L_t^{fsl} of an episode by Equation (7)
18: end for
19: for $(x_t^{C_l,1}, x_t^{C_l,2}) \epsilon C_l$ do
20: Calculate the L_t^{cl} of an episode by Equation (8)
21: end for
22: Return the L_t^{total} of an episode by Equation (3)
23: if episode $\%E_{test} == 0$
24: Predicting unlabeled samples with NN classifier
25: end if
26: end for

2) Novel Class Data: The IP dataset is obtained by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor, with 224 bands ranging from 0.4 to 2.5 μm. After removing the spectrum covering the water absorption range, 200 bands are retained. The image includes 145 × 145 pixels, and the spatial resolution is 20 m/pixel. The UP dataset was collected by the Reflective Optics Spectrographic Imaging System (ROSIS) sensor. It consists of 610 × 340 pixels with a spatial resolution of 1.3 m/pixel and 103 spectral bands ranging from 0.43 to 0.86 μm after 12 noisy channels being removed. The SA dataset was also obtained by AVIRIS at wavelengths of 0.4–2.5 μm. After abandoning water absorption bands, 204 bands were retained. The spatial size is 512 × 217, and the geometric resolution is 3.7 m/pixel. The WHU-Hi-HC dataset was collected by the 17-mm focal length Headwall Nano-Hyperspec imaging sensor installed on the V1 platform of the Leica Aibot X6 UAV. The UAV flight height is 250 m, and the image size is 1217 × 303 pixels, with a total of 274 bands. The spatial resolution of the UAV-borne HSI is about 0.109 m. Notably, since this dataset is obtained in the afternoon with low solar elevation angle, there are many shadow coverage areas in the image. Tables II–V detail the name of land cover classes and the number of samples of UP, IP, SA, and HC. Figs. 3–6 show the pseudocolor images and ground truth maps of HC, IP, UP, and SA.

TABLE II
NUMBER OF SAMPLES FOR THE UP DATASET

Class	Name	Number of Samples
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Sheets	1345
6	Bare soil	5029
7	Bitumen	1330
8	Bricks	3682
9	Shadow	947
Total		42776

TABLE III
NUMBER OF SAMPLES FOR THE IP DATASET

Class	Name	Number of Samples
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-tree	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93
Total		10249

B. Experimental Settings

To evaluate the performance of the proposed FSCF-SSL, relevant algorithms, including SVM, 3-D CNN (3DCNN) [16], SSRN [17], GIA-CFSL [33], DCFSL [31], and HFSL [32], are used for comparison. To ensure the fairness of the experiment, for some methods (GIA-CFSL and DCFSL), 200 samples are randomly selected from each base class for training to learn transferable knowledge. For the HFSL method and the proposed FSCF-SSL method, the training on base classes is performed using all the training sets of mini-ImageNet. For all comparison methods, 1–5 samples are selected from

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TABLE IV
NUMBER OF SAMPLES FOR THE SA DATASET

Class	Name	Number of Samples
1	Brocoli_green_weeds_1	2009
2	Brocoli_green_weeds_2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807
Total		54129

TABLE V
NUMBER OF SAMPLES FOR THE HC DATASET

Class	Name	Number of Samples
1	Strawberry	6710
2	Cowpea	3412
3	Soybean	1543
4	Sorghum	802
5	Water spinach	180
6	Watermelon	679
7	Greens	885
8	Trees	2696
9	Grass	1420
10	Red roof	1577
11	Gray roof	2536
12	Plastic	551
13	Bare soil	1367
14	Road	2784
15	Bright object	170
16	Water	11310
Total		38622

each novel class for training, and the remaining samples are used for testing. The workstation used in the experiment is configured as I9-10900X (3.7 GHz), 128 GB memory, and Nvidia GeForce RTX 3090 (24 GB), and the open-source software framework is PyTorch.

In our proposed FSCF-SSL, each sample is represented by a $33 \times 33 \times B$ data cube. For SA, UP, IP, and HC, the training episodes for base classes are 1100, 550, 900, and 1000, respectively, and the training episodes for novel classes are 1000. The dropout value for data augmentation in novel classes is 0.28, 0.26, 0.54, and 0.5, the testing episodes for novel classes are 20, 20, 50, and 50. The batch size for SSLGT and SSLCL is 128.

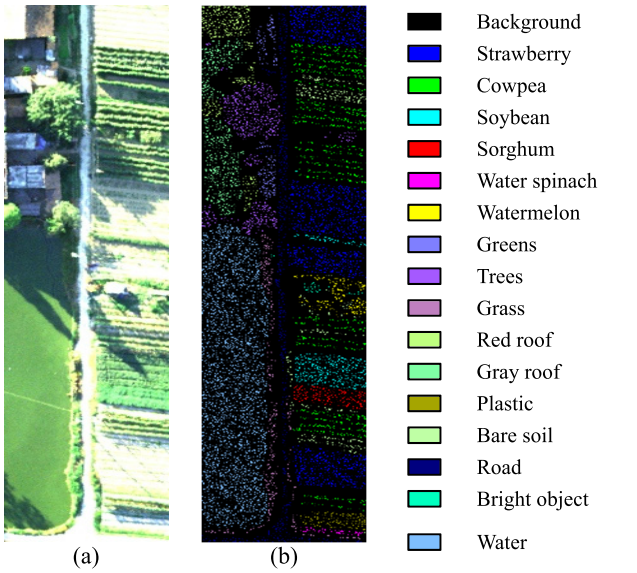

Fig. 3. Pseudocolor image and ground-truth map of HC. (a) Pseudocolor image of HC. (b) Ground-truth map of HC.

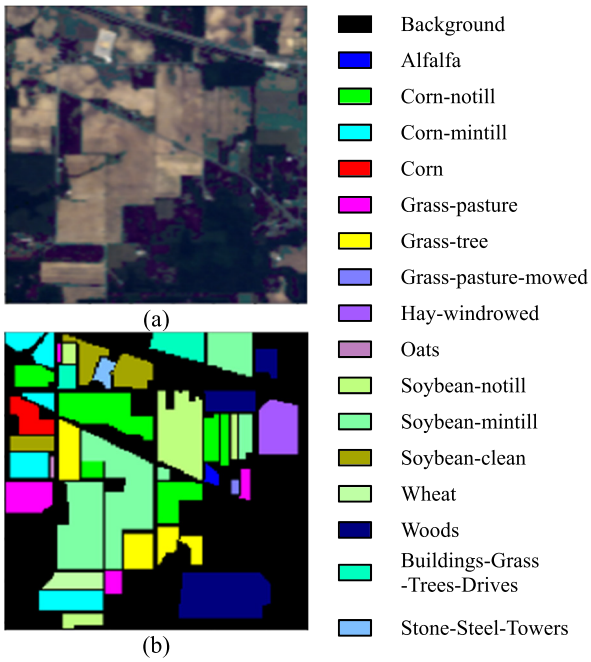

Fig. 4. Pseudocolor image and ground-truth map of IP. (a) Pseudocolor image of IP. (b) Ground-truth map of IP.

For FSL, each training episode can be regarded as a N-way, K-shot classification task, where N is the number of classes, and K is the number of samples selected in each class. Here, K is set to one, that is, each class selects a labeled sample to form a support set. The number of samples for each class in the query set is set to 19 because it is theoretically shown that the larger the number of samples, the better the simulation of the few samples class [30] in the testing dataset. In addition, our model is optimized by the Adam optimizer and the learning rate is set to 0.001. The structure of the feature extraction in FSCF-SSL is shown in detail in Table VI. N/A represents not applicable.

Overall classification accuracy (OA), average classification accuracy (AA), and kappa coefficient (Kappa) are used to evaluate the classification performance of all methods.

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TABLE VI
STRUCTURE OF FEATURE EXTRACTION

	Layer name	Output Shape	Filter size
Base class feature extraction	INPUT	33x33x3	N/A
	VGG16 (1-7)	4x4x256	N/A
	Conv 1	1x1x512	3x3,512
	FC 1	100	N/A
	FC 2	4	N/A
Spatial mapping	INPUT	33x33xB	N/A
	Conv 1	33x33x3	1x1,3
Spatial feature extraction	INPUT	33x33x3	N/A
	VGG16 (1-7)	4x4x256	N/A
	Conv 1	1x1x512	3x3,512
	FC	100	N/A
Spectral feature extraction	INPUT	band	N/A
	Conv 1	97x24	7x7,24
	Conv 2	97x24	7x7,24
	Conv 3	97x24	7x7,24
	Conv 4	1x128	(B - 7) // 2 + 1,128
	FC	100	N/A
Spectral-spatial feature fusion	INPUT	200	N/A
	FC 1	64	N/A
	FC 2	Num_class	N/A

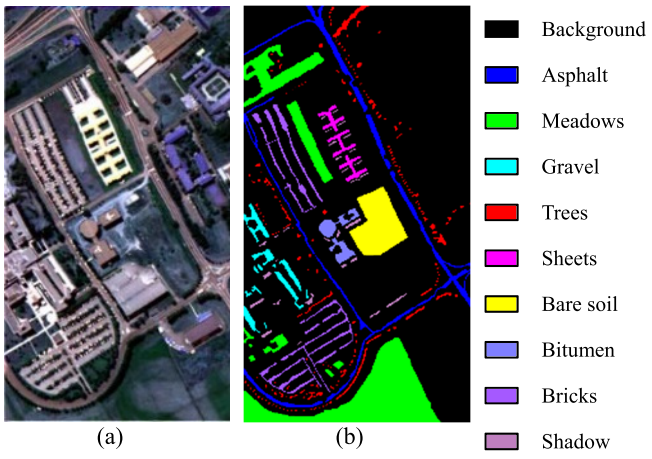

Fig. 5. Pseudocolor image and ground-truth map of UP. (a) Pseudocolor image of UP. (b) Ground-truth map of UP.

Fig. 6. Pseudocolor image and ground-truth map of SA. (a) Pseudocolor image of SA. (b) Ground-truth map of SA.

C. Experiment Results

For several supervised methods (SVM, 3DCNN, and SSRN), the training set can only be constructed by selecting samples from novel classes because the training and testing classes must be consistent. Specifically, the above method randomly selects five samples from novel classes as the training set and the remaining samples as the testing set. Notably, the kernel SVM nonlinearly maps data into a linearly separable high-dimensional feature space for classification, which only uses spectral information in HSI and does not consider spatial information. The 3DCNN method extracts deep spectral-spatial joint features to accurately classify HSI. SSRN uses spectral and spatial residual blocks to extract spectral-spatial features for classification.

GIA-CFSL, DCFSL, and HFSL belong to few-shot classification methods, which do not require the base and novel classes to have the same classes. GIA-CFSL is a cross-domain FSL framework based on graph information aggregation, which combines FSL with domain alignment based on graph information aggregation. The DCFSL method uses the domain adaptation method to solve the domain shift between the original dataset and the target dataset. The HFSL method uses the natural image dataset as the original dataset to learn rich texture and structure information to help HSI image classification.

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TABLE VII
CLASSIFICATION RESULTS (%) ON HC DATASET

Class	SVM	3DCNN	SSRN	GIA-CFSL	DCFSL	HFSL	FSCF-GT	FSCF-CL	FSCF-SSL
1	58.75	30.10	62.71	72.04	74.72	68.27	75.44	73.42	**78.14**
2	28.71	38.03	30.17	44.51	43.04	77.77	80.97	80.24	**81.59**
3	53.10	49.70	48.60	59.58	52.82	66.24	**73.90**	69.13	73.76
4	75.76	89.13	86.22	**90.88**	87.28	78.95	90.49	86.76	85.67
5	53.43	50.35	69.77	81.89	83.49	97.54	**98.91**	97.20	98.63
6	20.53	23.60	22.09	32.42	22.54	46.11	44.69	40.52	**50.52**
7	73.49	61.85	81.94	72.74	**83.11**	62.20	67.03	81.09	74.09
8	29.77	53.62	21.52	41.26	35.93	49.26	52.19	52.23	**54.87**
9	23.33	24.03	29.53	34.69	29.20	48.25	51.00	50.02	**53.67**
10	66.84	70.31	51.48	**72.54**	71.30	47.21	40.17	54.43	52.87
11	**84.51**	56.52	60.75	67.26	68.15	63.30	55.45	61.67	57.32
12	37.86	39.33	23.48	47.16	46.63	76.98	79.32	75.64	**83.11**
13	35.70	36.64	21.44	42.27	37.70	36.49	39.71	43.80	**45.23**
14	46.62	35.96	53.68	61.00	62.13	72.78	75.32	72.60	**80.30**
15	57.58	87.14	68.61	66.30	70.85	81.09	84.06	84.12	**85.94**
16	75.81	82.63	82.06	88.66	86.98	88.26	**95.79**	89.83	95.65
OA	57.58 ±4.10	54.73 ±3.84	57.87 ±3.25	67.70 ±3.70	66.64 ±4.04	70.85 ±2.82	75.07 ±3.60	73.73 ±3.88	**77.25 ±3.96**
AA	51.36 ±2.09	51.81 ±1.88	50.88 ±1.84	60.95 ±2.75	59.74 ±2.69	66.29 ±3.10	69.03 ±3.40	69.54 ±3.59	**71.96 ±3.10**
Kappa	52.00 ±4.29	49.01 ±3.98	51.95 ±3.34	62.89 ±3.92	61.71 ±4.35	66.43 ±3.01	71.01 ±4.10	69.73 ±4.29	**73.58 ±4.47**

TABLE VIII
CLASSIFICATION RESULTS (%) ON UP DATASET

Class	SVM	3DCNN	SSRN	GIA-CFSL	DCFSL	HFSL	FSCF-GT	FSCF-CL	FSCF-SSL
1	88.98	59.82	**91.84**	77.99	82.20	75.23	79.01	86.24	83.06
2	83.91	63.05	**95.13**	88.56	87.74	87.69	91.17	88.91	90.88
3	39.98	68.91	55.23	62.22	67.46	83.04	73.11	**85.56**	84.69
4	60.22	77.31	78.02	91.61	93.16	96.45	91.26	**97.05**	95.34
5	95.44	90.77	98.34	99.18	99.49	99.88	**99.96**	**99.96**	**99.96**
6	37.12	63.40	53.56	70.22	77.32	**93.38**	87.47	87.29	90.29
7	40.62	87.64	60.07	75.20	81.18	87.83	84.28	82.68	**91.00**
8	68.17	57.27	85.34	71.01	66.73	94.23	96.09	96.66	**97.41**
9	99.13	95.57	98.08	96.85	98.66	97.25	92.79	99.80	**99.95**
OA	64.12 ±4.55	65.74 ±1.77	76.26 ±5.78	82.28 ±1.91	83.65 ±1.77	87.98 ±3.39	88.49 ±2.80	89.78 ±2.73	**90.66 ±3.96**
AA	68.18 ±2.27	73.72 ±1.01	79.51 ±3.21	81.43 ±1.80	83.77 ±1.74	90.55 ±2.33	88.35 ±3.44	91.57 ±1.86	**92.51 ±1.45**
Kappa	55.59 ±4.64	57.37 ±1.97	70.56 ±6.69	76.84 ±2.27	78.70 ±2.01	84.50 ±4.15	84.93 ±3.56	86.69 ±3.40	**87.84 ±4.88**

Tables VII–X show the classification performance on HSI datasets (HC, UP, IP, and SA). For all classification methods, five samples are randomly selected from each novel class as supervised samples. To eliminate the impact of random sampling, all experiments were carried out ten times.

Table VII shows the classification results of all methods on the HC dataset. Compared with all comparison methods, the proposed FSCF-SSL method achieves the best classification results. For OA, AA, and Kappa, the FSCF-SSL method is 6.4%, 5.67%, and 7.15% higher than the HFSL method, and 9.55%, 11.01%, and 10.69% higher than the DCFSL method.

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TABLE IX
CLASSIFICATION RESULTS (%) ON IP DATASET

Class	SVM	3DCNN	SSRN	GIA-CFSL	DCFSL	HFSL	FSCF-GT	FSCF-CL	FSCF-SSL
1	72.20	95.12	18.38	91.71	95.37	99.27	99.27	99.76	**99.76**
2	34.27	37.70	**64.79**	41.12	43.26	58.11	54.51	59.26	61.16
3	39.18	19.77	27.65	51.94	57.95	68.25	71.28	63.56	**68.69**
4	50.34	32.51	26.97	79.44	80.60	88.02	92.89	91.38	**91.16**
5	69.75	**88.45**	80.76	74.31	72.91	76.17	75.00	72.70	70.59
6	66.36	73.65	86.87	81.57	**87.96**	81.94	81.38	82.36	82.84
7	89.13	81.82	32.24	99.13	99.57	**100**	99.57	99.57	99.57
8	68.73	53.35	**100**	90.15	86.26	99.75	99.92	94.12	98.65
9	86.67	**100**	57.69	94.67	99.33	**100**	100	100	**100**
10	37.49	41.35	59.69	60.42	62.44	59.72	65.02	59.15	**62.81**
11	33.96	66.71	70.87	60.54	62.75	68.22	73.07	75.11	**75.42**
12	31.43	37.40	45.00	45.31	48.72	67.74	62.40	66.11	**67.87**
13	86.50	85.71	88.29	97.30	**99.35**	97.95	98.55	99.55	99.30
14	62.93	62.57	**97.18**	78.04	85.40	93.12	90.36	93.13	96.07
15	28.08	56.42	36.64	69.06	66.69	88.77	91.44	90.89	**92.10**
16	90.91	90.36	60.98	98.52	97.61	95.57	96.70	96.36	97.16
OA	45.85 ±2.44	54.76 ±0.03	61.36 ±0.49	63.99 ±3.07	66.81 ±2.37	74.16 ±4.04	75.07 ±2.91	75.26 ±3.39	**77.02 ±3.37**
AA	59.24 ±1.36	63.93 ±0.02	59.75 ±0.20	75.83 ±2.44	77.89 ±0.86	83.91 ±1.86	84.46 ±1.94	83.94 ±1.96	**85.20 ±2.17**
Kappa	39.68 ±2.48	48.72 ±0.03	56.91 ±0.48	59.43 ±3.17	62.64 ±0.86	70.85 ±4.31	71.84 ±3.17	71.93 ±3.64	**73.96 ±3.70**

TABLE X
CLASSIFICATION RESULTS (%) ON SA DATASET

Class	SVM	3DCNN	SSRN	GIA-CFSL	DCFSL	HFSL	FSCF-GT	FSCF-CL	FSCF-SSL
1	97.57	95.29	97.55	99.56	99.40	99.93	97.18	99.81	**100**
2	87.43	97.20	98.97	98.51	**99.76**	94.13	95.64	96.68	97.11
3	82.95	91.45	92.47	89.96	91.96	93.79	94.55	93.45	**97.42**
4	99.11	97.31	96.50	98.99	99.55	98.53	99.59	99.68	**99.73**
5	94.29	91.24	94.20	87.10	92.70	96.75	95.01	95.90	**97.05**
6	98.36	98.80	99.28	98.64	99.52	99.14	98.91	**99.66**	**99.66**
7	94.39	99.69	**99.98**	96.77	98.88	97.27	94.65	94.96	94.33
8	59.99	66.40	**86.90**	76.78	74.57	64.94	78.71	80.66	80.54
9	96.09	96.25	**99.64**	98.24	99.59	99.05	98.74	99.61	99.11
10	71.45	70.72	92.01	79.95	86.42	92.14	91.47	94.91	**96.32**
11	91.25	93.15	95.86	97.05	96.61	**99.39**	98.72	99.11	98.84
12	97.22	99.65	99.15	99.31	**99.93**	99.19	98.37	98.52	98.95
13	97.30	92.63	89.24	97.77	99.30	98.22	**99.35**	98.63	98.87
14	91.84	93.56	95.15	97.93	**98.85**	97.46	98.68	96.42	97.61
15	60.52	68.02	55.97	74.51	75.38	**79.41**	77.12	79.23	77.29
16	81.45	81.41	**98.91**	87.68	92.22	93.70	92.15	96.68	98.74
OA	80.71 ±2.75	84.20 ±2.62	86.39 ±2.68	88.29 ±1.16	89.34 ±2.19	87.93 ±2.92	90.15 ±2.27	91.50 ±2.45	**91.55 ±2.16**
AA	87.58 ±1.84	89.56 ±1.79	93.24 ±1.29	92.42 ±0.97	94.04 ±1.14	93.94 ±1.16	94.30 ±1.36	95.24 ±1.72	**95.72 ±1.69**
Kappa	78.61 ±3.0	82.46 ±2.90	84.95 ±2.90	86.98 ±1.27	88.17 ±2.40	86.64 ±3.16	89.04 ±2.53	90.54 ±2.74	**90.60 ±2.41**

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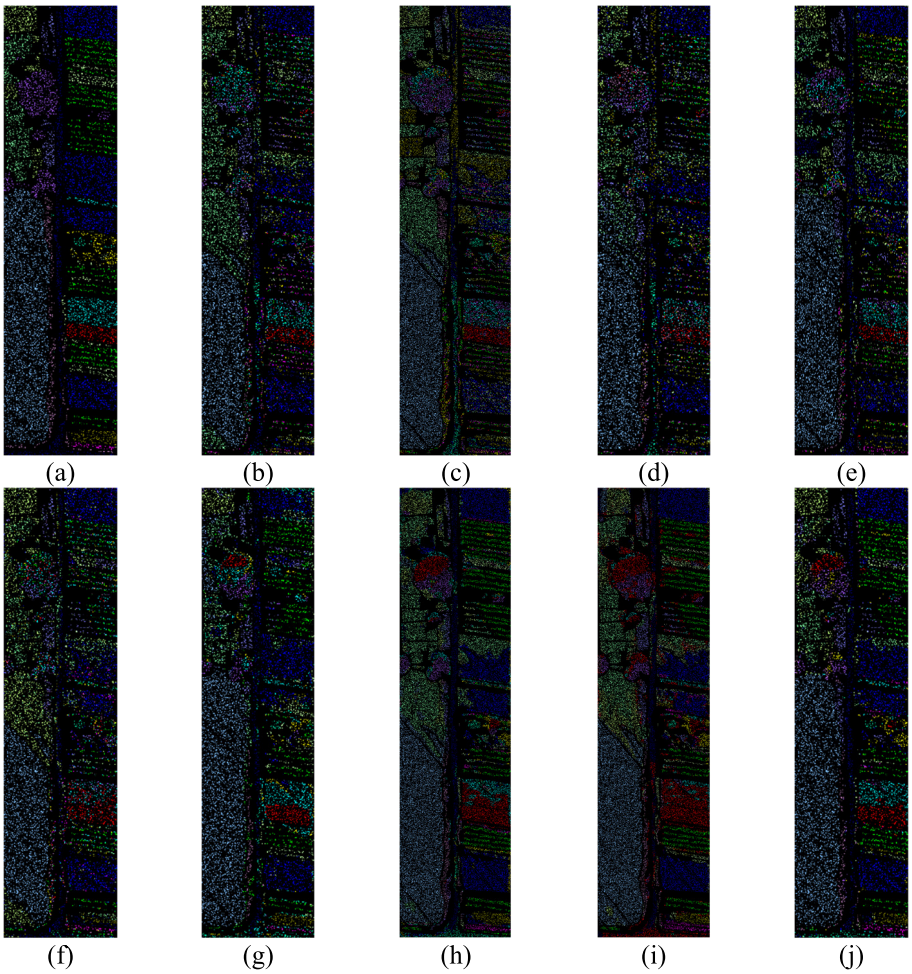

Fig. 7. Classification maps obtained by different methods on HC. (a) Ground-truth, (b) SVM, (c) 3DCNN, (d) SSRN, (e) GIA-FSL, (f) DCFSL, (g) HFSL, (h) FSCF-GT, (i) FSCF-CL, and (j) FSCF-SSL.

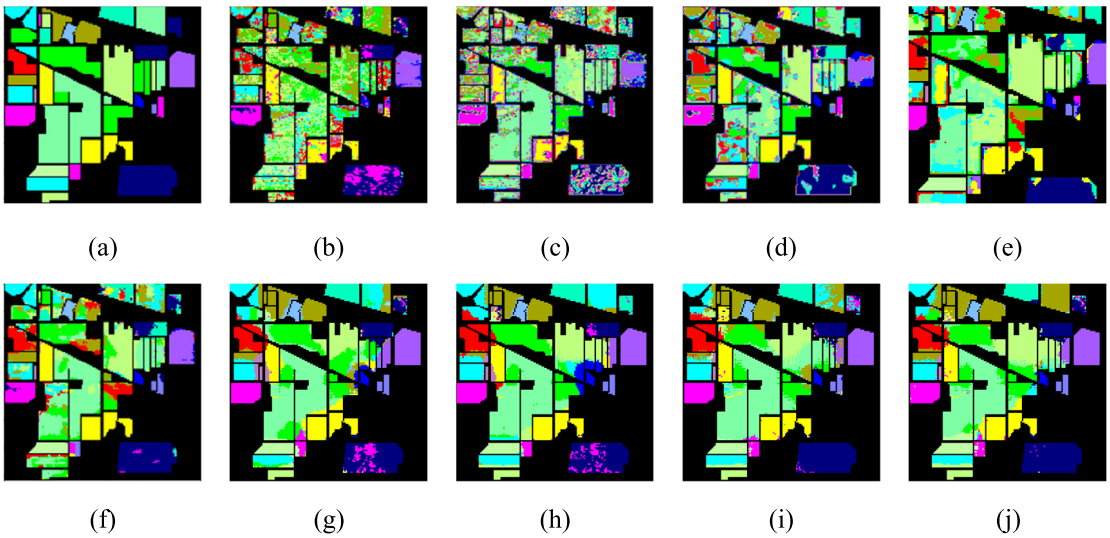

Fig. 8. Classification maps obtained by different methods on IP. (a) Ground-truth, (b) SVM, (c) 3DCNN, (d) SSRN, (e) GIA-FSL, (f) DCFSL, (g) HFSL, (h) FSCF-GT, (i) FSCF-CL, and (j) FSCF-SSL.

In particular, FSCF-SSL ranks first in nine categories, among which the accuracies of categories 1, 7, and 13 are more than 8% higher than that of the HFSL method, which fully proved the effectiveness of FSCF-SSL. In addition, compared with SVM, 3DCNN, and SSRN, the FSL method can obtain higher OA, AA, and Kappa because the FSL method can fully learn from the base class and make more use of the knowledge of novel class classification.

Table VIII shows the classification performance of the proposed FSCF-SSL and other comparison methods on the

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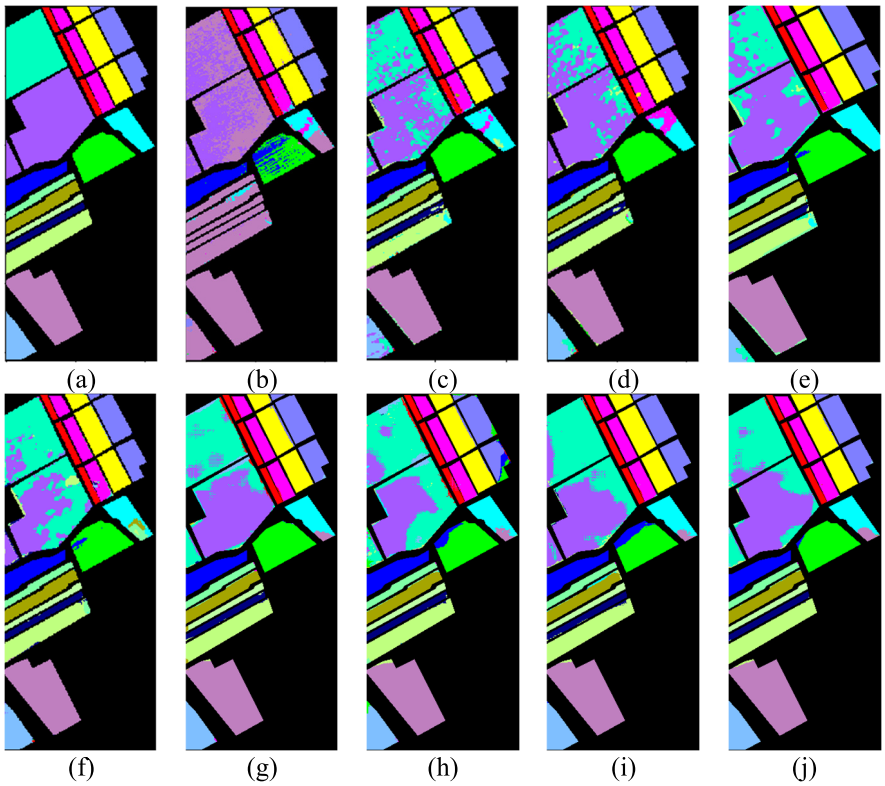

Fig. 9. Classification maps obtained by different methods on SA. (a) Ground-truth, (b) SVM, (c) 3DCNN, (d) SSRN, (e) GIA-FSL, (f) DCFSL, (g) HFSL, (h) FSCF-GT, (i) FSCF-CL, and (j) FSCF-SSL.

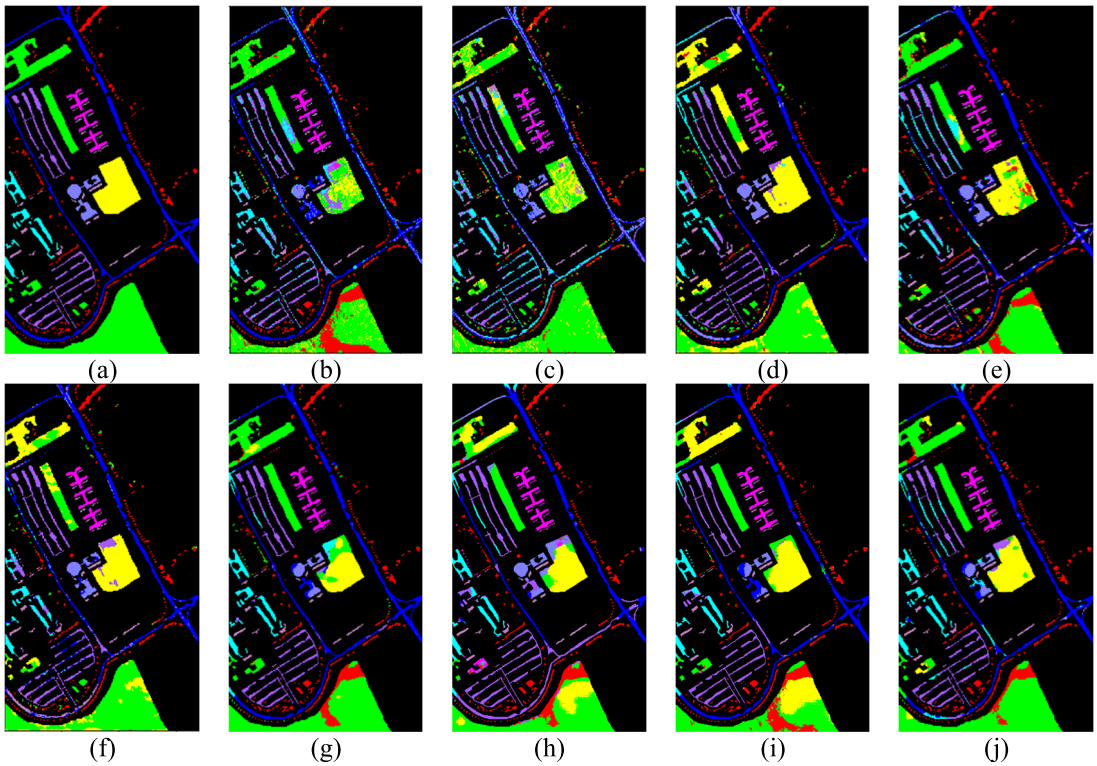

Fig. 10. Classification maps obtained by different methods on UP. (a) Ground-truth, (b) SVM, (c) 3DCNN, (d) SSRN, (e) GIA-FSL, (f) DCFSL, (g) HFSL, (h) FSCF-GT, (i) FSCF-CL, and (j) FSCF-SSL.

UP dataset. It is obvious that the OA, AA, and Kappa of FSCF-SSL are higher than those of other comparison methods. The OA, AA, and Kappa from the FSCF-SSL are 2.68%, 1.96%, and 3.34% higher than the HFSL, respectively, and 7.01%, 8.74%, and 9.14% higher than the DCFSL, which indicates that FSCF-SSL can extract more

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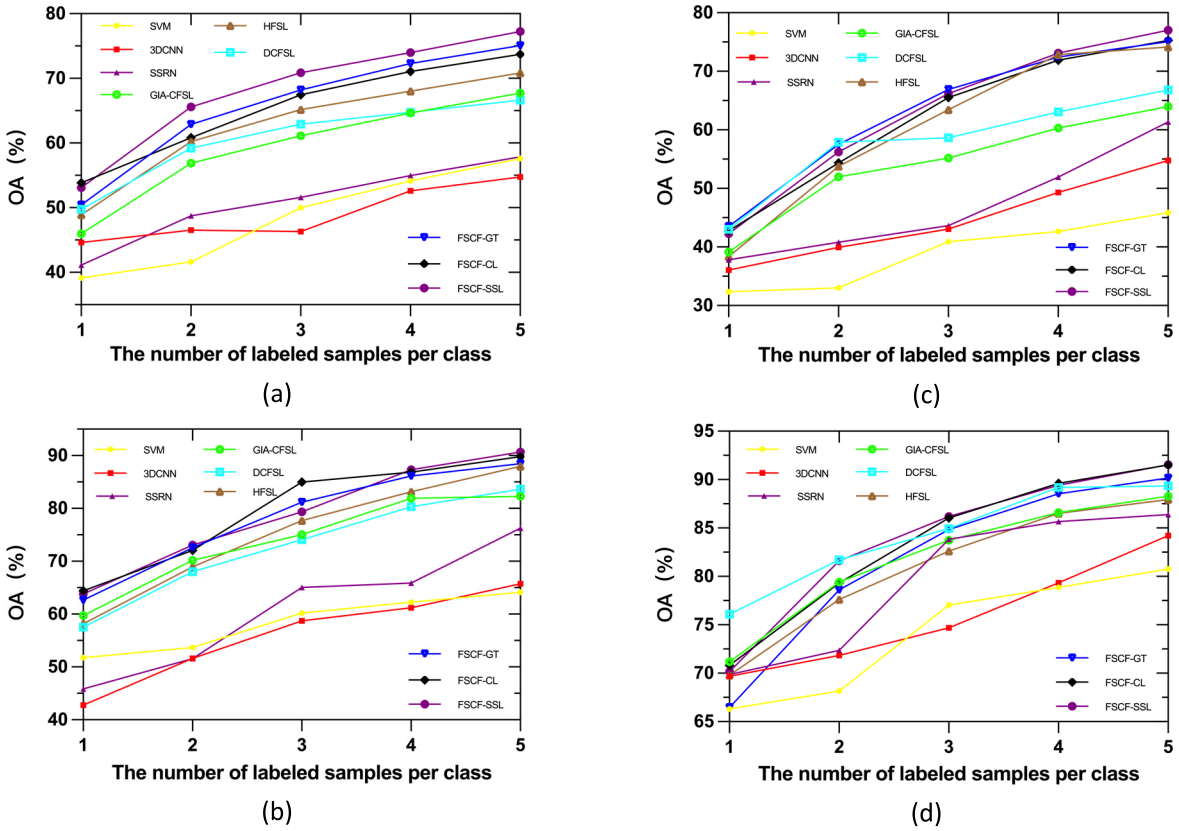

Fig. 11. Classification results of all methods under different number of labeled samples. (a) HC, (b) UP, (c) IP, and (d) SA.

TABLE XI
ABLATION COMPARISON OF EACH VARIANT OF FSCF-SSL

methods	IP	UP	SA	HC
FSCF	74.03	87.94	89.84	72.68
FSCF-GT	75.07	88.49	90.15	75.07
FSCF-CL	75.26	89.78	91.50	73.73
FSCF-SSL	77.02	90.66	91.55	77.25

discriminative features and thus improve the classification performance.

Table IX shows the classification performance of the proposed FSCF-SSL and other comparison methods on the IP dataset. The OA, AA, and Kappa generated by FSCF-SSL are 2.86%, 1.29%, and 3.11 % higher than the HFSL, and 10.21%, 7.31%, and 11.32% higher than the DCFSL, respectively. This is mainly because the FSCF-SSL method can learn richer spatial structure and texture information from base classes, thereby helping to classify novel classes.

Table X shows the classification accuracy of the SA dataset. It can be clearly seen that the proposed FSCF-SSL achieves the highest classification performance. In particular, the OA of the FSCF-SSL method is at least 2% higher than the DCFSL method and ranks first in six categories. Experimental results show that the high-quality category-invariant features learned by SSLCL are beneficial to the classification of novel classes.

To compare the classification of each algorithm more intuitively, Figs. 7–10 show the mapping maps for different novel class data with only five labeled samples per class. As can be seen from that, the classification map generated by the proposed FSCF-SSL method is most like the ground truth and the edge of different classes in the classification map is clearer. It shows that the FSCF-SLCL method can extract discriminative interclass features.

To evaluate the classification performance of the proposed FSCF-SSL under different numbers of labeled samples, 1–5 samples are randomly selected from novel classes to construct the training set, and each experiment is repeated ten times. Fig. 11 shows the results of all methods on different datasets.

As can be seen from Fig. 11, the classification accuracies of the proposed FSCF-SSL method are ahead of all the comparison methods on the HC dataset and the UP dataset. In particular, the experimental results of FSCF-SSL on the HC dataset have been greatly improved compared to the HFSL method. Specifically, it has increased by at least 4% in the case of each different labeled sample. For IP and SA datasets, the proposed FSCF-SSL method is not optimal when the number of samples is equal to 1 and 2. However, the accuracies of the method are the best among all methods when the number of samples was increased to 3, 4, and 5. One possible reason is that the IP and SA datasets contain many fine-grained feature categories, which undoubtedly makes classification more difficult. More importantly, in our approach, there is a greater difference between natural images used as base classes

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TABLE XII
COMPUTATION TIME (S), FLOPS (M), AND PARAMETERS (K) ON EACH DATASET WITH ALL METHODS

	Methods	SVM	3DCNN	SSRN	GIA-CFSL	DCFSL	HFSL	FSCF-GT	FSCF-CL	FSCF-SSL
UP	Training time	0.09	3.96	6.94	2857.91	1874.20	2824.68	7787.35	8730.70	11650.00
	Testing time	0.16	1.51	3.81	3.92	5.58	10.70	9.01	10.24	10.77
	FLOPS	-	0.14	14.56	27719.00	46.20	398.81	597.75	798.56	997.50
	#Params	-	3.92	198.42	857.12	4259.29	6126.50	6126.91	6126.50	6126.91
HC	Training time	0.14	7.02	11.05	4420.65	1998.20	7122.46	10063.15	14619.86	14480.32
	Testing time	0.41	1.58	3.72	4.56	4.26	17.40	18.26	19.25	19.99
	FLOPS	-	0.39	39.20	27720.96	47.58	400.36	599.29	803.19	1002.13
	#Params	-	17.64	459.71	875.10	4276.39	6388.58	6388.99	6388.58	6388.99
SA	Training time	0.12	7.08	10.98	4116.33	1899.68	6476.91	16839.27	18803.58	18825.05
	Testing time	0.44	2.04	5.16	5.23	5.19	21.86	21.88	21.88	21.88
	FLOPS	-	0.29	29.05	27720.39	47.01	399.72	598.66	801.29	1000.23
	#Params	-	13.16	352.19	868.10	4269.39	6280.85	6281.26	6280.85	6281.26
IP	Training time	0.13	7.04	11.20	4620.14	1833.29	6026.73	6744.20	7933.03	10405.05
	Testing time	0.10	0.4	0.93	1.46	0.94	4.13	3.92	4.06	4.38
	FLOPS	-	0.28	28.47	27720.36	46.98	399.69	598.62	801.18	1000.12
	#Params	-	12.91	346.05	867.70	4268.99	6274.70	6275.10	6274.70	6275.10

and HSIs used as novel classes and the HSI used by the novel class, and thus too few novel class labeled samples (such as 1 or 2) may not be sufficient to learn more knowledge that is beneficial for novel class classification.

D. Ablation Analysis

To further show the effectiveness of SSLGT used in the base class training and the SSLCL used in the novel class training, four methods are evaluated on the basis of FSL on the two domains: 1) using FSL only on base classes and novel classes (FSCF); 2) adding SSLGT to FSCF (FSCF-GT); 3) adding SSLCL to FSCF (FSCF-CL); and 4) adding SSLGT and SSLCL to FSCF (FSCF-SSL). Different ablation models are listed in Table XI. It can be seen that FSCF-GT can extract low-level features from natural images that can better represent different orientations, and can promote the learning of spatial meta-knowledge suitable for HSIs, making OA increase by 1.04%, 0.55%, 0.31%, and 2.39% on IP, UP, SA, and HC, respectively. FSCF-CL can obtain high-quality category-invariant features to help classification, so that OA is increased by at least 1% on four different datasets. FSCF-SSL combined with all modules provides excellent results, with an OA increase of 4.57% on HC and at least 1.5% on other datasets. Obviously, all modules improve the classification performance of the four datasets.

E. Computational Complexity

The computational efficiency of all methods is evaluated by analyzing the number of parameters, the number of floating-point operations (FLOPS), training time, and prediction time. The details are shown in Table XII. For some methods (SVM, 3DCNN, and SSRN), the base class data and the novel class data from the same dataset, the training time is the time to train the model with base classes, and the testing time is the time to test novel classes. For these methods (GIA-CFSL, DCFSL, HFSL, FSCF-GT, FSCF-CL, and FSCF-SSL), the base class data and the novel class data from different datasets, the training time includes model training time using base classes, model transfer time, model fine-tuning time using novel class labeled data, and testing time is the time to predict unlabeled samples in novel classes. As can be seen from Table XII, methods of the base classes and new classes from different datasets have longer training times and more FLOPS than methods from the same dataset, which indicates that the cost of transfer learning is higher. Our proposed methods (FSFC-GT, FSFC-CL, and FSFC-SSL) achieve better classification performance despite longer training time.

V. CONCLUSION

In this article, a novel method called FSCF-SSL has been proposed to solve FSL classification for HSIs. Unlike most current FSL-based HSI classification methods, FSCF-SSL focuses on meta-knowledge transfer between natural images and HSIs. Considering that the objects of image patches for each pixel in HSIs have diverse orientations in the spatial domain, we design an SSLGT to extract low-level features from natural images, which can better represent diverse orientations from natural images. By performing FSL and SSLGT on natural images, the model can learn the transferable spatial meta-knowledge more suitable for HSIs. In addition, we devise an SSLCL to fully explore the few annotated data in HSIs, which can learn the high-quality category-invariant features contained in the data itself by sharpening the category distribution of each sample and scattering the predictions across the samples. The model can learn more discriminative individual

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knowledge from HSIs by performing SSLCL and FSL on HSIs. Extensive experiments on four datasets demonstrated that the proposed method offers a significant improvement over state-of-the-art methods.

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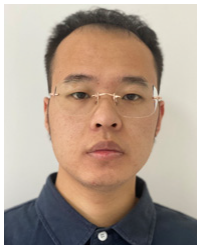

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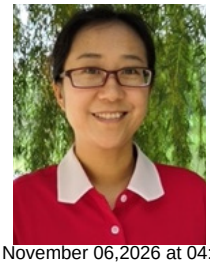

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